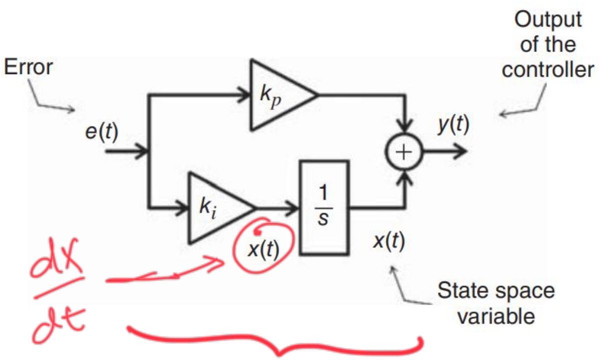
Problem 2

For the MATLAB code below, make changes such that you feed the RL load with a half-bridge single-phase converter controlled by hysteresis.

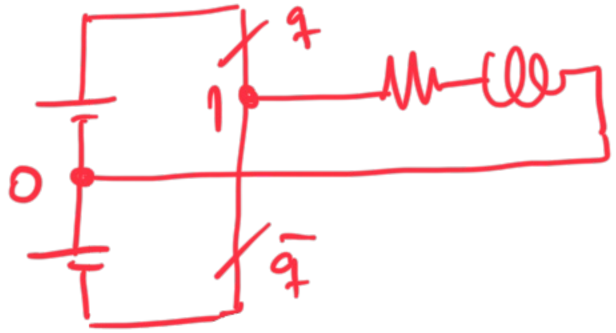
Problem 3

Repeat Problem 2 by changing the hysteresis control to PI controller, see below.

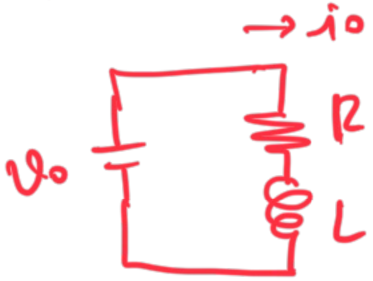
```
%Simulation of a RL load connected to a Controlled-
Voltage-Source
R=5;    %Resistance of the RL load
L=0.05; %Inductance of the RL load
io=0;      %state variable (load current)
vo=100;    %input variable (load voltage)
h=1e-4;    %step size
j=0;       %variable used to storage the outcomes
t=0;       %initial value of the digital time
tmax=1;    %maximum simulation time

while t<=tmax, %begining of the simulation loop
      t = t + h; %time incrementation
      io = io + (vo-R*io)*h/L;  %state space variable 1
      % storage of the variables
      j = j + 1;
      time(j)=t;
      current(j)=io;
      voltage(j)=vo;
end
   figure(1),plot(time,voltage)
   figure(2),plot(time,current)
```

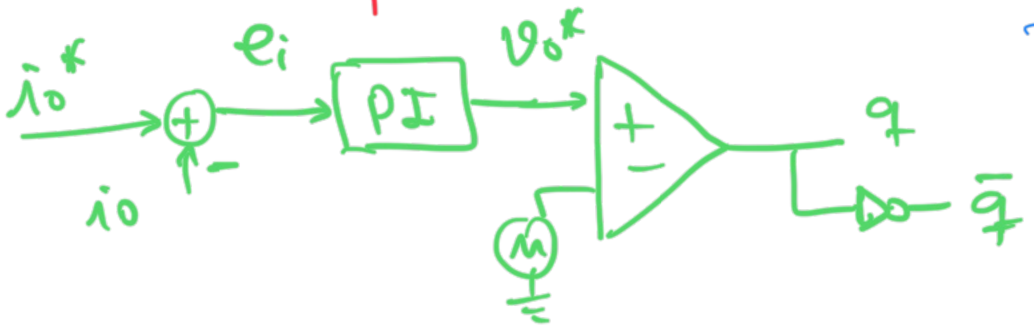


② CHANGE THE CODE TO IMPLEMENT THIS:



① THIS CODE IS IMPLEMENTING THIS:





ANALOG IMPLEMENTATION

③ THE CODE WILL LOOK LIKE THIS:

PI CONTROLLER:

$$\frac{dx}{dt} = ki\, e_i \Rightarrow x = x + ki\, e_i \cdot h$$

$y = x + kp \cdot e_i$; WHERE $y = v_o^*$

```
⋮
io* = 5;  % REFERENCE CURRENT

WHILE t ≤ tmax,
    t = t + h;
    ei = io* - io;
    x = x + ki·ei·h;  % PI CONT.
    vo* = x + kp·ei;  % PI CONT.

    IF (t > tpwm)
        tpwm = tpwm + hpwm;
        CONT = 0;
        TAL = (vo*/Vdc + 1/2)·hpwm;

    CONT = CONT + h;

    IF (CONT < TAL)
        q = 1
    ELSE
        q = 0

    vo = (2q1 - 1)·Vdc/2
    io = io + (vo - R·io)·h/L;
END
```

DIGITAL IMPLEMENTATION

AN ALTERNATIVE SOLUTION WOULD BE TO CONSIDER THE HALF-BRIDGE AS AN IDEAL ELEMENT, THEN $v_0^* = v_0$

$$\text{WHILE } t \leq t_{max};$$

$$t = t + h;$$

$$e_i = i_0^* - i_0;$$

$$x = x + k_i e_i \cdot h;$$

$$v_0^* = x + k_p \cdot e_i;$$

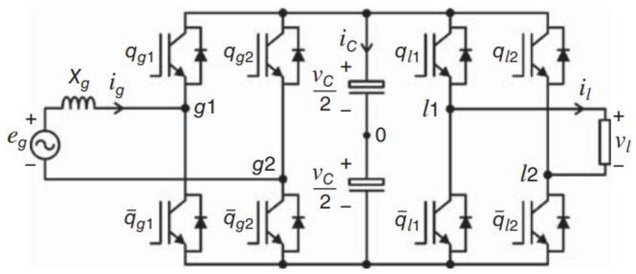
$$v_0 = v_0^*;$$

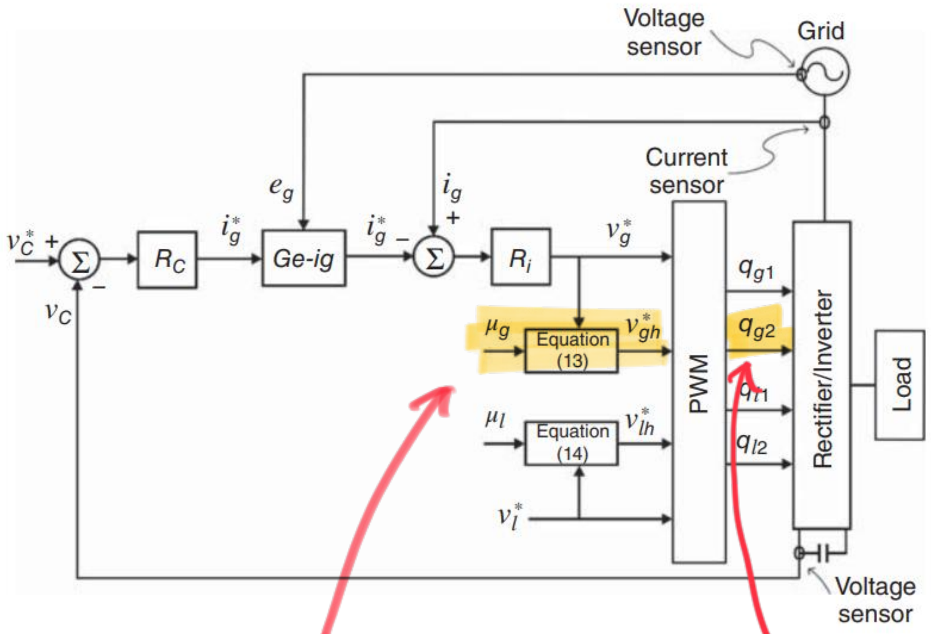
$$i_0 = i_0 + (v_0 - R \cdot i_0) \cdot h / L;$$

$$\vdots$$

Problem 4

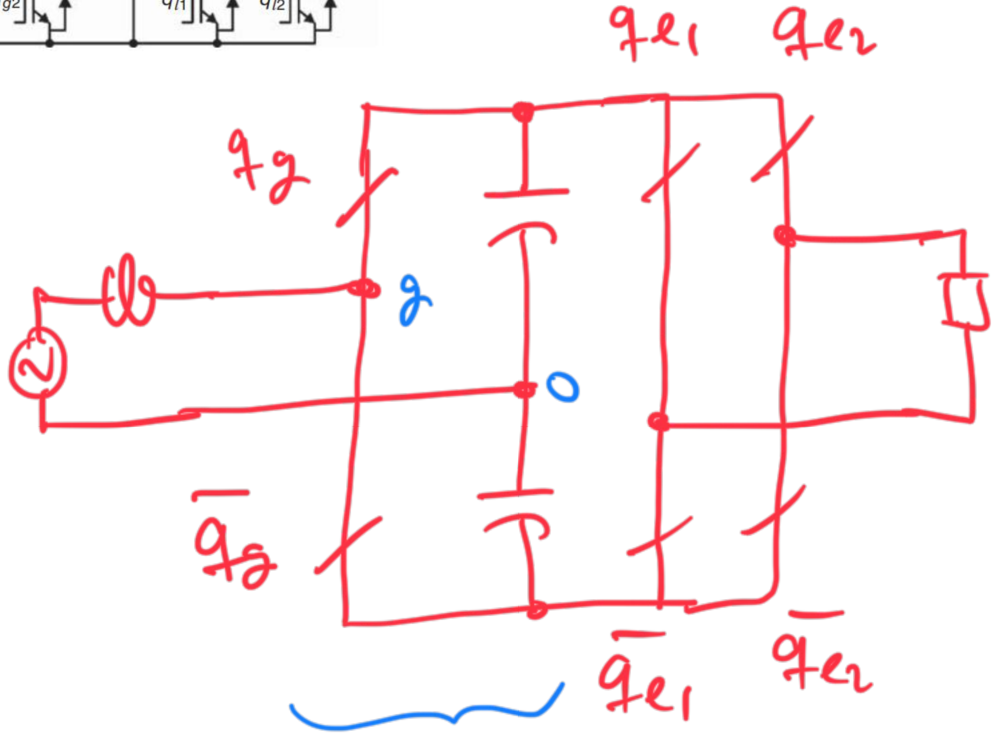
Below is presented a single-phase B2B topology and its control strategy. Propose changes in the block diagram aside if the input is a half-bridge rectifier, everything else remains the same.



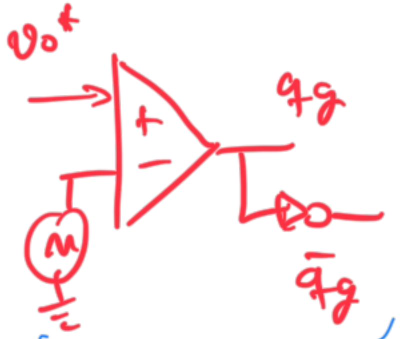


SINCE THERE IS ONLY ONE LEG AT THE INPUT, THERE IS NO OPPORTUNITY TO USE CPWM

NO q_{g2}



HALF-BRIDGE



ANALOG

$$T_g = \left(\frac{v_g^*}{V_{dc}} + \frac{1}{2}\right) T_s$$

DIGITAL